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Introduction: The following presentation is about the INNER, LEFT and FULL OUTER JOINS SQL syntax made on data bricks. Screenshots with explanations and table will be shown to see the null values where applicable.

Run selected (1000) ✓ 2 minutes ago (19s) workspace. default New SQL editor: ON ☆

```
select * from `laboritory`.`default`.`shows` limit 100;
-- Q1 INNER JOIN between a users table and subscribers table and column (user_id) as the PK.
SELECT c.User AS user_id,
      c._c1 AS user_name,
      s.Subscriptions AS subscription_id,
      s._c2 AS plan_id,
      s._c3 AS start_date
FROM `laboritory`.`default`.`users` c
INNER JOIN `laboritory`.`default`.`subscriptions` s
      ON c.User = s._c1;
```

Add parameter

Table +

	^A _C user_id	^A _C user_name	^A _C subscription_id	^A _C plan_id	^A _C start_date
1	1	Nomvula	501	10	1/18/26
2	2	David	502	11	2/1/26
3	1	Nomvula	503	12	3/10/26
4	3	Anele	505	13	4/5/26

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```
--Q2 INNER JOIN between subscriptions table and plans table using the (plan_id) column as the PK
SELECT s.Subscriptions AS subscription_id,
       s._c1 AS user_id,
       p.Plans AS plan_id,
       p._c2 AS monthly_price
FROM `laboratory`.`default`.`subscriptions` s
INNER JOIN `laboratory`.`default`.`plans` p
ON s._c2 = p.Plans;
```

Add parameter

Table +

	^B _C subscription_id	^B _C user_id	^B _C plan_id	^B _C monthly_price
1	Subscriptions_id	User_id	Plan_id	Monthly price
2	501	1	10	79
3	502	2	11	129
4	503	1	12	199
5	504	6	11	129
6	505	3	13	249

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```
--Q3 INNER JOIN between viewing session table and shows table using show_id column as the primary key.
SELECT v.`Viewing sessions` AS session_id,
       v._c1 AS user_id,
       v._c2 AS show_id,
       v._c3 AS watch_minutes,
       s._c1 AS show_title,
       s._c2 AS genre
FROM `laboratory`.`default`.`viewing_session` v
INNER JOIN `laboratory`.`default`.`shows` s
ON v._c2 = s.shows;
```

Add parameter

Table +

	^B _C session_id	^B _C user_id	^B _C show_id	^B _C watch_minutes	^B _C show_title	^B _C genre
1	session_id	user_id	show_id	watch_minutes	show_title	Genre
2	901	1	701	45	Comedy hour	Comedy
3	902	2	703	30	Tech Tale	Documentary
4	903	1	702	60	Crime time	Drama
5	904	7	701	20	Comedy hour	Comedy

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```
--Q4 INNER JOIN between users table and viewing session table using the user_id column as the primary key.
SELECT s.User AS user_id,
       s._c1 AS user_name,
       s._c2 AS country,
       v.`Viewing sessions` AS session_id,
       v._c2 AS show_id,
       v._c3 AS watch_minutes
FROM `laboratory`.`default`.`users` s
INNER JOIN `laboratory`.`default`.`viewing_session` v
ON s.User = v._c1;
```

Add parameter

Table +

	^A _C user_id	^A _C user_name	^A _C country	^A _C session_id	^A _C show_id	^A _C watch_minutes
1	user_id	user_name	country	session_id	show_id	watch_minutes
2	1	Nomvula	Johannesburg	901	701	45
3	2	David	Cape Town	902	703	30
4	1	Nomvula	Johannesburg	903	702	60
5	3	Anele	Durban	905	705	60

```
--Q5 INNER JOIN between three columns users, plans and subscriptions table, user_id(PK) on users table and (FK) on subscription table, Plan_id (PK) on the subscription table and (FK) on the plans table.
SELECT s.User AS user_id,
       s._c1 AS user_name,
       s._c2 AS country,
       b.Subscriptions AS subscription_id,
       b._c3 AS start_date,
       p.Plans AS plan_id,
       p._c1 AS plan_name,
       p._c2 AS monthly_price
FROM `laboratory`.`default`.`users` s
INNER JOIN `laboratory`.`default`.`subscriptions` b
ON s.User = b._c1
INNER JOIN `laboratory`.`default`.`plans` p
ON b._c2 = p.Plans;
```

Add parameter

Table +

	^A _C user_id	^A _C user_name	^A _C country	^A _C subscription_id	^A _C start_date	^A _C plan_id	^A _C plan_name	^A _C monthly_price
1	1	Nomvula	Johannesburg	501	1/18/26	10	Basic	79
2	2	David	Cape Town	502	2/1/26	11	Standard	129
3	1	Nomvula	Johannesburg	503	3/10/26	12	Premium	199
4	3	Anele	Durban	505	4/5/26	13	Family	249

```
-- Q6 LEFT JOIN subscription table to users table using the user_id column as the PK, null values are on subscriptio_id and start date columns
SELECT s.User AS user_id,
       s._c1 AS user_name,
       s._c2 AS country,
       b.Subscriptions AS subscription_id,
       b._c3 AS start_date
FROM `laboritory`.`default`.`users` s
LEFT JOIN `laboritory`.`default`.`subscriptions` b
ON s.User = b._c1;
```

add parameter

Table ▾ +

	^A _C user_id	^A _C user_name	^A _C country	^A _C subscription_id	^A _C start_date
1	user_id	user_name	country	null	null
2	1	Nomvula	Johannesburg	503	3/10/26
3	2	David	Cape Town	502	2/1/26
4	3	Anele	Durban	505	4/5/26
5	4	Kabelo	Pretoria	null	null
6	5	Lerato	Port Elizabeth	null	null
7	1	Nomvula	Johannesburg	501	1/18/26

Snipping Tool

Screenshot copied to clipboard
Automatically saved to screenshots folder.

```
--Q7 LEFT JOIN the subscriptions table to the plans table using the plan_id column as the Primary key. null values are on subscription_id, user_id and plan_id.
SELECT s.Subscriptions AS subscription_id,
       s._c1 AS user_id,
       s._c2 AS plan_id,
       p._c1 AS plan_name
FROM `laboritory`.`default`.`plans` p
LEFT JOIN `laboritory`.`default`.`subscriptions` s
ON p.Plans = s._c2;
```

add parameter

Table ▾ +

	^A _C subscription_id	^A _C user_id	^A _C plan_id	^A _C plan_name
1	Subscriptions_id	User_id	Plan_id	Plan_name
2	501	1	10	Basic
3	504	6	11	Standard
4	503	1	12	Premium
5	505	3	13	Family
6	null	null	null	Mobile
7	502	2	11	Standard

```
--Q8 LEFT JOIN the viewing session to the shows table using the show_id column as the primary key. null values are on the sessions_id and watch_minutes columns.
SELECT s.shows AS show_id,
s._c1 AS show_title,
s._c2 AS genre,
v.`Viewing sessions` AS session_id,
v._c3 AS watch_minutes
FROM `laboratory`.`default`.`shows` s
LEFT JOIN `laboratory`.`default`.`viewing_session` v
ON s.shows = v._c2;
```

Add parameter

	^B _C show_id	^B _C show_title	^B _C genre	^B _C session_id	^B _C watch_minutes
1	show_id	show_title	Genre	session_id	watch_minutes
2	701	Comedy hour	Comedy	904	20
3	702	Crime time	Drama	903	60
4	703	Tech Tale	Documentary	902	30
5	704	Cooking lab	Lifestyle	null	null
6	706	Wild earth	documentary	null	null
7	701	Comedy hour	Comedy	901	45

```
-- Q9 LEFT JOIN users table to the viewing session using the user_id column as the primary key. null values are on the user_name and country columns.
SELECT v.`Viewing sessions` AS session_id,
...v._c1 AS user_id,
...v._c2 AS show_id,
...v._c3 AS watch_minutes,
...s._c1 AS user_name,
...s._c2 AS country
FROM `laboratory`.`default`.`viewing_session` v
... LEFT JOIN `laboratory`.`default`.`users` s
... ON s.User = v._c1;
```

Add parameter

	^B _C session_id	^B _C user_id	^B _C show_id	^B _C watch_minutes	^B _C user_name	^B _C country
1	session_id	user_id	show_id	watch_minutes	user_name	country
2	901	1	701	45	Normvula	Johannesburg
3	902	2	703	30	David	Cape Town
4	903	1	702	60	Normvula	Johannesburg
5	904	7	701	20	null	null
6	905	3	705	60	Anele	Durban

```
--Q10 LEFT JOIN two tables subscriptions and plans to users table user_id as the primary key on the users table and foreing key on the
subscription table, plan_id as the primary key on the subscription table and foreign key on the plan table. null values are on the
subscription_id, plan_id, plan_name and monthly_price columns.
SELECT u.User AS user_id,
u._c1 AS user_name,
u._c2 AS country,
s.Subscriptions AS subscription_id,
s._c2 AS plan_id,
p._c1 AS plan_name,
p._c2 AS monthly_price
✓ FROM `laboritory`.`default`.`users` u
LEFT JOIN `laboritory`.`default`.`subscriptions` s
ON u.User = s._c1
LEFT JOIN `laboritory`.`default`.`plans` p
ON s._c2 = p.Plans;
```

Add parameter

	user_id	user_name	country	subscription_id	plan_id	plan_name	monthly_price
1	user_id	user_name	country	null	null	null	null
2	1	Nomvula	Johannesburg	503	12	Premium	199
3	2	David	Cape Town	502	11	Standard	129
4	3	Anele	Durban	505	13	Family	249

```
--Q11 FULL OUTER JOIN subscription table to the users table. all columns have null values
SELECT u.User AS user_id,
u._c1 AS user_name,
u._c2 AS country,
s.Subscriptions AS subscription_id,
s._c2 AS plan_id,
s._c3 AS start_date
✓ FROM `laboritory`.`default`.`users` u
FULL OUTER JOIN `laboritory`.`default`.`subscriptions` s
ON u.User = s._c1;
```

Add parameter

	user_id	user_name	country	subscription_id	plan_id	start_date
1	3	Anele	Durban	505	13	4/5/26
2	null	null	null	504	11	3/20/26
3	2	David	Cape Town	502	11	2/1/26
4	null	null	null	Subscriptions_id	Plan_id	Start_date
5	1	Nomvula	Johannesburg	501	10	1/18/26
6	1	Nomvula	Johannesburg	503	12	3/10/26
7	5	Lerato	Port Elizabeth	null	null	null

```
--Q12 FULL OUTER JOIN the table subscription to the plans table using the plan_id column as the primary key.

SELECT p.plans AS plan_id,
       p._c1 AS plan_name,
       p._c2 AS monthly_plan,
       s._c1 AS subscription_id,
       s._c2 AS user_id,
       s._c3 AS start_date
FROM `laboratory`.`default`.`plans` p
FULL OUTER JOIN `laboratory`.`default`.`subscriptions` s
ON p.plans = s._c1;

-- Q13 FULL OUTER JOIN subscription table to the users table
```

Add parameter

Table ▼ +

	A _C plan_id	A _C plan_name	A _C monthly_plan	A _C subscription_id	A _C user_id	A _C start_date
1	14	Mobile	59	null	null	null
2	12	Premium	199	null	null	null
3	10	Basic	79	null	null	null
4	13	Family	249	null	null	null
5	11	Standard	129	null	null	null
6	Plan_id	Plan_name	Monthly price	null	null	null
7	null	null	null	3	13	4/5/26

```
--Q13 FULL OUTER JOIN viewing sessions table to the shows table using the show_id column as the primary key.

SELECT s.shows AS show_id,
       s._c1 AS show_title,
       s._c2 AS genre,
       v.`Viewing sessions` AS session_id,
       v._c1 AS user_id,
       v._c2 AS viewing_show_id,
       v._c3 AS watch_minutes
FROM `laboratory`.`default`.`shows` s
FULL OUTER JOIN `laboratory`.`default`.`viewing_session` v
ON s.shows = v._c2;
```

Add parameter

Table ▼ +

	A _C show_id	A _C show_title	A _C genre	A _C session_id	A _C user_id	A _C viewing_show_id	A _C watch_minutes
1	702	Crime time	Drama	903	1	702	60
2	null	null	null	905	3	705	60
3	701	Comedy hour	Comedy	901	1	701	45
4	701	Comedy hour	Comedy	904	7	701	20
5	show_id	show_title	Genre	session_id	user_id	show_id	watch_minutes
6	703	Tech Tale	Documentary	902	2	703	30
7	706	Wild earth	documentary	null	null	null	null

```
--Q14 FULL OUTER JOIN viewing session table on the users table user_id column being the primary key.
SELECT u.User AS user_id,
u._c1 AS user_name,
u._c2 AS country,
v.`Viewing sessions` AS session_id,
v._c1 AS viewing_user_id,
v._c2 AS show_id,
v._c3 AS watch_minutes
FROM `laboratory`.`default`.`users` u
FULL OUTER JOIN `laboratory`.`default`.`viewing_session` v
ON u.User = v._c1;
```

Add parameter

	^B _C user_id	^B _C user_name	^B _C country	^B _C session_id	^B _C viewing_user_id	^B _C show_id	^B _C watch_minutes
1	3	Anele	Durban	905	3	705	60
2	2	David	Cape Town	902	2	703	30
3	1	Nomvula	Johannesburg	901	1	701	45
4	1	Nomvula	Johannesburg	903	1	702	60
5	user_id	user_name	country	session_id	user_id	show_id	watch_minutes
6	null	null	null	904	7	701	20
7	5	Lerato	Port Elizabeth	null	null	null	null
~	~	~	~	~	~	~	~

```
-- Q15 FULL OUTER JOIN users, subscription and plans tables, user_id(PK) on users table and (FK) on subscription table, plan_id(PK) on subscription table and (FK) on the plans table.
```

```
SELECT u.User AS user_id,
u._c1 AS username,
u._c2 AS country,
s.Subscriptions AS subscription_id,
s._c1 AS subscription_user_id,
s._c2 AS subscription_plan_id,
s._c3 AS start_date,
p.Plans AS plan_id,
p._c1 AS plan_name,
p._c2 AS monthly_plan
FROM `laboratory`.`default`.`users` u
FULL OUTER JOIN `laboratory`.`default`.`subscriptions` s
ON u.User = s._c1
FULL OUTER JOIN `laboratory`.`default`.`plans` p
ON s._c2 = p.Plans;
```

```
-- Q15 FULL OUTER JOIN users, subscription and plans tables, user_id(PK) on users table and (FK) on subscription table, plan_id(PK) on subscription table and (FK) on the plans table.
```

Add parameter

	^B _C user_id	^B _C username	^B _C country	^B _C subscription_id	^B _C subscription_user_id	^B _C subscription_plan_id	^B _C start_date	^B _C plan_id
1	1	Nomvula	Johannesburg	503	1	12	3/10/26	12
2	1	Nomvula	Johannesburg	501	1	10	1/18/26	10
3	3	Anele	Durban	505	3	13	4/5/26	13
4	5	Lerato	Port Elizabeth	null	null	null	null	null
5	user_id	user_name	country	null	null	null	null	null
6	4	Kabelo	Pretoria	504	null	null	null	null
7	null	null	null	504	6	11	3/20/26	11
8	2	David	Cape Town	502	2	11	2/1/26	11
9	null	null	null	Subscriptions_id	User_id	Plan_id	Start_date	Plan_id
10	null	null	null	null	null	null	null	14

Conclusion: shows the code of INNER, LEFT INNER and FULL OUTER JOINS.